Roll No. | | | | | | | |

SET-2

End Semester Examination 2025

Name of the Course: M.Tech. Biomedical Engineering
Name of the Paper: Nanotechnology in Medicine
Time: 3 Hours

Semester: I
Paper Code: **BEM 143**
Maximum Marks: 100

Note:-

(iv) All questions are compulsory.
(v) Answer any two sub questions among a, b & c in each main question.
(vi) Each sub question carries 10 marks.

Q1	(20 marks)	
(a)	Discuss the role of **intermolecular forces** in determining the thermodynamic behavior of many-particle systems.	
(b)	Compare and contrast phenomenological potentials, pseudo-potentials, and many-body potentials.	**CO1**
(c)	Explain the concept of a **potential energy surface (PES)** in the context of molecular interactions.	
Q2	(20 marks)	
(a)	What are **carbon nanotubes (CNTs)**? Discuss their types, basis of classification, and properties (mechanical, electrical, and thermal).	
(b)	Briefly discuss the effect of nano-dimensions on **electrical, magnetic, and optical** properties of materials.	**CO2**
(c)	(i) What are different forces involved at nano-domain? (ii) How size of the nanoparticles can be determined?	
Q3	(20 marks)	
(a)	Briefly describe **co-precipitation** and **microemulsion** techniques of fabricating nanomaterials along with their general mechanism, reaction characteristics and advantages.	
(b)	What is **molecular beam epitaxy (MBE)**? With a suitable sketch, describe detailed process of fabricating crystalline thin film using MBE technique.	**CO3**
(c)	Give an account of the discovery of CNTs and briefly explain **CVD** and **Laser Ablation** methods of synthesizing CNTs.	
Q4	(20 marks)	
(a)	What is the principle of **X-ray diffraction**? Elaborate on working and applications of X-ray diffraction in nanotechnology.	
(b)	Give a detailed account of principle, working and applications of **X-ray absorption spectroscopy**.	**CO4**
(c)	Briefly discuss about key components, working, and applications of **Scanning Electron Microscope (SEM)**.	
Q5	(20 marks)	
(a)	Briefly discuss the concept of **field emission** and **shielding** in molecular electronics.	
(b)	Write a short note on the various **polymeric drug delivery** systems.	**CO5**
(c)	What is **quantum dots technology**? Discuss their classification, and applications in bio-imaging and bio-sensors.	